



K. J. Somaiya College of Engineering, Mumbai-77
(A Constituent College of Somaiya Vidyavihar University)
Department of Computer Engineering

Batch: B3 Roll No.: 1611124

Experiment No. 8

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Implementation of N-Queen Problem using Backtracking Algorithm

Objective: To learn the Backtracking strategy of problem solving for 8-Queens problem

CO to be achieved:

Sr. No	Objective
CO 1	Compare and demonstrate the efficiency of algorithms using asymptotic complexity notations.
CO 2	Analyze and solve problems for divide and conquer strategy, greedy method, dynamic programming approach and backtracking and branch & bound policies.

Books/ Journals/ Websites referred:

1. Ellis horowitz, Sarataj Sahni, S.Rajsekaran," Fundamentals of computer algorithm", University Press
2. T.H.Cormen ,C.E.Leiserson,R.L.Rivest and C.Stein," Introduction to algortihmts",2nd Edition ,MIT press/McGraw Hill,2001
3. <http://www.math.utah.edu/~alfeld/queens/queens.html>
4. <http://www-isl.ece.arizona.edu/ece175/assignments275/assignment4a/Solving%208%20queen%20problem.pdf>
5. http://www.slideshare.net/Tech_MX/8-queens-problem-using-back-tracking
6. <http://www.mathcs.emory.edu/~cheung/Courses/170.2010/Syllabus/Backtrac king/8queens.html>
7. <http://www.geeksforgeeks.org/backtracking-set-3-n-queen-problem/>
8. <http://www.hbmeyer.de/backtrack/achtdamen/eight.htm>



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Pre Lab/ Prior Concepts:

Data structures, Concepts of algorithm analysis

Historical Profile:

The **N-Queens puzzle** is the problem of placing N queens on an N×N chessboard so that no two queens attack each other. Thus, a solution requires that no two queens share the same row, column, or diagonal.

New Concepts to be learned:

Application of algorithmic design strategy to any problem, Backtracking method of problem-solving Vs other methods of problem solving, 8- Queens problem and its applications.

Algorithm N Queens Problem: -

```
void NQueens(int k, int n)
```

```
// Using backtracking, this procedure prints all possible placements of n queens on an n X n chessboard so that they are nonattacking.
```

```
{   for (int i=1; i<=n; i++)
    {
        if (Place(k, i))
        {
            x[k] = i;
            if (k==n)
                for (int j=1; j<=n; j++)      Print x[j] ;
            else NQueens(k+1, n);
        }
    }
}
```

```
Boolean Place(int k, int i)
```

```
// Returns true if a queen can be placed in kth row and ith column. Otherwise it returns false.
```

```
// x[] is a global array whose first (k-1) values have been set. abs(r) returns absolute value of r.
```

```
{
for (int j=1; j < k; j++)
    if ((x[j] == i) // Two in the same column
        || (abs(x[j]-i) == abs(j-k))) // or in the same diagonal
        return(false);
return(true);
}
```



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Example 8-Queens Problem:

The eight queens puzzle is the problem of placing eight chess queens on an 8×8 chessboard so that no two queens threaten each other i.e. no two queens share the same row, column, or diagonal.

Solution Using Backtracking Approach:

The idea is to place queens one by one in different columns, starting from the leftmost column. When we place a queen in a column, we check for clashes with already placed queens. In the current column, if we find a row for which there is no clash, we mark this row and column as part of the solution. If we do not find such a row due to clashes then we backtrack and return false.

State Space tree for N-Queens (Solution):

Implementation (Code):

OUTPUT:

Algorithm:



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Analysis of Backtracking solution:

CONCLUSION:

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